Supplement

Table S1: hDNA and aDNA extraction protocols used for type specimens

a) Bones, teeth: Protocols must take into account that the DNA is enclosed in hard calcified cavities.

Bones, teeth, shells	**Advantages**	**Disadvantages**	**Examples of type specimen studies**
Pioneering ancient DNA protocols with decontamination & decalcification of bone, guanidinium thiocyanate extraction with phenol-chloroform purification (Russell and Sambrook 2001) and silica-in-solution purification (Höss and Pääbo 1993)	manual, inexpensive	Complex, time-consuming manual work	Krings et al.(1997; Mammalia), adapted for tissue by Kirchmann et al. (2009; Aves), Hofreiter et al. (2001; Mammalia), Dalebout et al. (2014; Mammalia)
Different, specific extraction buffers, purification using GuSCN/silica suspension by Rohland et al. (2004), Hofreiter et al. (2004)	manual, inexpensive	Complex, time-consuming manual work	Scherz et al. (2020; Amphibia)
Early protocol combining decontamination & decalcification of bone with DNeasy Tissue Kit (Qiagen), Kearney & Stuart (2004)	silica-column based kit combination saves time	small DNA fragments are not retained	Stuart and Fritz (2008; Testudines)
Silica-in-solution protocol to reduce further DNA damage, Rohland and Hofreiter (2007)	scalable, PCR inhibitors removed	Requires substantial time for buffer & silica suspension preparation	Olsen et al. (2016; Mammalia), Li et al. (2015; Pisces)
Silica-based column, Rohland et al. (2010)	faster than in-solution silica protocol	Single company producing specialised columns	Li et al. (2015; Pisces)
Subfossil bone protocol targeting fragmented, short single-stranded DNA via modified use of a MinElute silica column in a custom-built apparatus, Dabney et al. (2013)	DNA yield of short fragments is much improved	Complex manual preparation, expensive custom-built apparatus	Kehlmaier et al. (2017; Testudines), Scherz et al. 2020; Amphibia), Vershinina et al. (2020; Mammalia), Kehlmaier et al. (2023; Testudines)

b) Tissue, wet and dry: DNA extraction of tissue from museum specimens (plants, fungi and animals) has a much more diverse set of approaches.

Wet tissue	**Advantages**	**Disadvantages**	**Examples of type specimen studies**
Dabney et al. (2013) Extraction protocol adapted for tissue.	Improved DNA yield of short fragments	Complex and time-consuming manual work, expensive custom-built apparatus	McGuire et al. (2018; Lepidosauria), Rancilhac et al. (2020; Amphibia), Boubli et al. (2018; Mammalia)
Dabney et al. Extraction protocol adapted for tissue with proteinase-K lysis from Sambrook & Russell (2001) by Straube et al. (2021a)	Improved DNA yield of short fragments	Complex and time-consuming manual work, expensive custom-built apparatus	Straube et al. (2021b; Pisces), Vences et al. (2021; Amphibia), Agne et al. (2022; Pisces)
PacBio ultra-low library protocol adapted with alternative whole genome amplification polymerase, Bein et al. (2025)	Minimal DNA input with high sequencing performance, reduced PCR bias with alternative whole genome amplification polymerase.	Whole genome amplification step introduces risk of PCR bias.	none yet
Phenol/chloroform extraction for well-preserved frozen tissue in ethanol	Well-proven and de facto standard method for high-quality genomes	Irritating chemicals	Schöneberg et al. (2023; Lepidosauria)
Sera-Mag library cleanup SpeedBeads adapted for DNA extraction, Rázuri-Gonzales et al. (2022)	Inexpensive	Tricky pipetting, requires magnetic apparatus	Mahony et al. (2022; Amphibia), Heckenhauer et al. (2023; Insecta)
Hot alkaline lysis gDNA extraction from formalin-fixed archival tissues (V.2), Hahn et al. (2022)	Good DNA yield, robust protocol, good potential for difficult samples	Low-throughput, requires specific devices, 2-day protocol	Hahn et al. (2022)
Dry tissue	**Advantages**	**Disadvantages**	**Examples of type specimen studies**
CTAB extraction protocol for plants, Doyle and Doyle (1987) by Rogers and Bendich (1985) and partly adapted by Porebski et al. (1997)	Well-proven	Toxic chemicals, complex and time-consuming manual work	Larsson and Jacobsson (2004; Basidiomycota), Andreasen et al. (2009; Streptophyta), Hughey and Gabrielson (2012; Rhodophyta), Hosaka and Uno (2013; Basidiomycota), Erpenbeck et al. (2016; Porifera), Richards et al. (2021b; Rhodophyta), Contreras-Ortiz et al. (2019; Angiosperms), Hughey et al. (2019; Chlorophyta), Richards et al. (2021a; Rhodophyta)

Mercaptoethanol extraction buffer, Dellaporta et al. (1983), followed by chloroform extraction step and DNA precipitation, Hughey et al. (2001)	Well-proven	Toxic chemicals, complex and time-consuming manual work	(Rhodophyta: Hughey et al. 2002, 2014; Gabrielson et al. 2011; Hind et al. 2014; Mateo-Cid et al. 2014; Hernandez-Kantun et al. 2015; Maneveldt et al. 2020; Calderon et al. 2021; Peña et al. 2021; Richards et al. 2021a; Schipper et al. 2023; Wade et al. 2023) (Chlorophyta: Hughey et al. 2024)
DNAzol reagent (Gibco BRL/Life Technologies, Gaithersburg, MD, U.S.A.) combined with chaotropic agent guanidine thiocyanate and detergent mixture for lysis, permits selective precipitation of DNA from a cell lysate by Junqueira et al. (2002)	Best-performing of three tested methods, better performance than chelex and phenol/chloroform for old specimens	Toxic chemicals, complex and time-consuming manual work	Maddison and Cooper (2014; Insecta)
Immersion of entire beetles followed by phenol/chloroform (Sambrook et al. 1989) extraction by Gilbert et al. (2007)	Minimally destructive, well-proven	Toxic chemicals, complex and time-consuming manual work	Maddison and Cooper (2014; Insecta)
Silica-based glass fibre filtration (GF) plates in 96-well format for automation, (Ivanova et al. 2006; Dewaard et al. 2008)	Inexpensive, automation-friendly	Targets HMW DNA/long fragments, degraded small fragments lost	(Insecta: Vaglia et al. 2008; Rougerie et al. 2012; Mutanen et al. 2015; Speidel et al. 2015; Prosser et al. 2016; Hausmann et al. 2017)
Mechanical disruption of spores using single acid-washed glass bead (Frommlet and Iglesias-Rodríguez 2008)	Minimally destructive, no DNA cleanup, direct to PCR	Delicate procedure	Janik et al. (2020; Myxomycetes)
sbeadex forensic kit (LGC), Hundsdoerfer and Kitching (2010)	Purifies DNA of different sizes using magnetic beads, adaptable for formalin-fixed wet tissue	Special magnet equipment needed, tricky pipetting	Stuckas and Fritz (2021; Lepidosauria, formalin-fixed), Mende and Hundsdoerfer (2013; Insecta), Fritz et al. (2014; Testudines), Petzold et al. (2014; Testudines), Hundsdoerfer et al.(2017; Insecta), Hundsdoerfer and Kitching (2017; Insecta), Kehlmaier et al. (2020; Lepidosauria, formalin-fixed), Schweizer et al. (2020; Aves)
Chelex X-100 beads (BioRad) (Junqueira et al. 2002)	Cheap	DNA is not purified	Shumskaya et al. (2023; Basidiomycota), Ferencova et al. (2017, lichens)

E.Z.N.A. Plant DNA DS Mini Kit (Omega Biotek) (Kistenich et al. 2019)	Fast, easy, well-proven	Targets HMW DNA/long fragments, degraded small fragments lost	Leavitt et al. (2019; lichen)
Oligo-nucleotide and PCR-cleanup kit (NEB) (Patzold et al. 2020), Qiagen QIAquick PCR purification kit (Knyshov et al. 2019a)	Fast, targets both small/degraded DNA fragments and large fragments	Special magnet equipment needed, tricky pipetting	Patzold et al. (2021; Insecta), Knyshov et al. (2019b; Insecta)
GEN-IAL All-tissue DNA-Kit (GEN-IAL GmbH, Troisdorf, Germany)	Cheap, universally applicable	Somewhat time-consuming	(Stelbrink et al. 2019; Mollusca), Wilke et al. (2023; Mollusca)
QIAGEN DNA isolation kits (e.g., DNeasy Blood & Tissue kit, Genomic-tip 20/G, QIAamp DNA Investigator Kit, QiAmp Mini Kit, QIAamp DNA Micro Kit) (Qiagen, West Sussex, UK or Hilden, Germany) used for dry plant, fungal and animal tissue	Fast, easy, well-proven	Expensive, targets HMW DNA/long fragments, but can also work well for smaller fragments	Ronikier et al. (2022; Myxomycetes, Amoebozoa), Ernst et al. (2021; Amphibia), Kirschel et al. (2018; Aves), Collinson et al. (2018; Aves), Shumskaya et al. (2023; Basidiomycota), Taylor et al. (2017; Chlorophyta), Puillandre et al. (2011; Crustacea), (Insecta: Strutzenberger et al. 2012; Mammalia: Cappellini et al. 2014; Price et al. 2015; Mantellatto et al. 2021; Mayer et al. 2021; Twort et al. 2021; Castañeda-Rico et al. 2022), Vieira et al. (2016; Ochrophyta), (Pisces: Silva et al. 2019; Sullivan et al. 2022), Erpenbeck et al. (2016; Porifera), (Testudines: Praschag et al. 2008; Kehlmaier et al. 2019), (Rhodophyta: Hernandez-Kantun et al. 2015, 2016; Suzuki et al. 2016; Peña et al. 2021; Gabrielson et al. 2023a, 2023b)
QiaQuick PCR purification kit (Qiagen, West Sussex, UK or Hilden, Germany)	Fast, targets both small/degraded DNA fragments and large fragments	Expensive	Kirschel et al. (2018; Aves)
NucleoSpin Plant II Kit (Macherey-Nagel, Düren, Germany)	Fast, easy, well-proven	Targets HMW DNA/long fragments, degraded small fragments lost	Hausmann et al. (2009; Insecta), Chomicki and Renner (2015; Angiosperms), Jeong et al. (2019; Rhodophyta), Maneveldt et al. (2020; Rhodophyta)

QuickExtract Plant DNA Extraction Solution (Epicentre, Madison, WI, USA)	Extremely fast, simple, non-toxic reagents, no mechanical procedures or centrifugation needed	DNA is not purified	Hanyuda and Kawi (2018; Chlorophyta)
Sorbitol washing complex homogenate for improved DNA extractions (Jones et al. 2021)	Removal of polyphenols and polysaccharides in plant and fungi samples	Toxic chemicals	none yet

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